

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Education is an important instrument for individual development, social transformation and national development. Traditionally, formal education has largely depended on physical interaction between teachers and students within classrooms, lecture halls and other designated learning environments. In this conventional arrangement, tutors deliver lessons directly to students, students ask questions, participate in discussions, receive feedback and complete academic activities under the supervision of their instructors. Although the conventional classroom remains important, developments in Information and Communication Technology (ICT) have significantly changed how teaching and learning can be delivered.

The growth of the Internet, mobile computing, cloud computing and web technologies has provided educational institutions with alternative methods of delivering instructional content and facilitating communication. One of these methods is electronic learning, commonly referred to as e-learning. E-learning involves the use of electronic and digital technologies to support teaching, learning, communication, assessment and access to educational resources. It may include learning management systems, online classrooms, discussion forums, electronic materials, digital assessments, messaging systems and synchronous virtual meetings. The increasing importance of e-learning in higher education has been documented by (Baig et al. 2022), whose systematic review found that research on e-learning has increasingly focused on adoption, acceptance, readiness, user experiences and the challenges associated with implementation.

One major advantage of e-learning is that it can reduce the dependence of teaching and learning on physical location. A tutor and a student may participate in an academic activity even when they are not physically present in the same classroom. This characteristic is particularly relevant to higher education, where students may have different schedules, geographical locations and academic responsibilities. E-learning can also support flexible access to course materials and communication between learners and instructors. However, flexibility alone does not guarantee effective learning. The quality of interaction between tutors and students remains an important component of the online learning experience.

Interaction is particularly important because teaching is not simply the transmission of information from an instructor to a learner. Effective learning involves explanation, questioning, feedback, clarification, discussion, collaboration and assessment. (Moore, 1989) identified learner's instructor, learner's content and learner's interaction as important dimensions of distance education. These forms of interaction remain relevant to contemporary online learning because students require opportunities to communicate with instructors, engage with learning resources and collaborate with other learners.

The development of online meeting technologies has made synchronous interaction possible. Synchronous e-learning allows tutors and students to communicate at approximately the same time through audio, video, text chat, screen sharing and other collaborative tools. Unlike purely asynchronous learning, where participants may interact at different times, synchronous meetings can create a learning experience that more closely resembles the immediacy of a physical classroom. This can be particularly useful when a tutor needs to explain difficult concepts, conduct demonstrations, supervise discussions, answer questions or provide immediate feedback. The importance of interaction is supported by empirical research. (Miao et al. 2022), in a study involving 354 undergraduate students at a Chinese public university, examined teacher-student interaction, student-student interaction and social presence in online learning. Their findings showed that teacher-student interaction and student-student interaction directly influenced social presence and learning engagement, while social presence also had a direct relationship with learning engagement. This indicates that an online learning system should not merely provide digital content; it should also facilitate meaningful interaction among the major participants in the learning process.

Student engagement is another important issue in online learning. Engagement refers broadly to the extent to which students participate behaviorally, cognitively and emotionally in learning activities. (Nkomo et al. 2021), in a systematic review of literature on student engagement with digital technologies, emphasized that digital technologies can influence how students participate in learning, although engagement depends on the nature and design of the learning environment. Similarly, (Ahmadi et al. 2023) reviewed 32 studies and identified indicators of engagement in learning management systems including system usage, access to course materials, assignments, assessments, messaging and forum participation.

The role of the tutor is therefore important in maintaining an effective online learning environment. Online systems should provide mechanisms through which tutors can communicate with students, organize meetings, provide learning resources, monitor participation and give feedback. (Xu et al. 2020) found that teacher facilitation in online discussion environments was associated with higher behavioral and cognitive engagement among students, demonstrating the importance of active instructor involvement rather than merely providing students with access to an online platform.

The COVID-19 pandemic further demonstrated the importance of systems capable of supporting remote education. The closure of educational institutions forced many institutions to move rapidly from face-to-face teaching to online and virtual learning. Although this transition demonstrated the potential of technology to support educational continuity, it also exposed weaknesses in infrastructure, digital skills, communication and system design. Research conducted in Nigeria illustrates these challenges clearly.

For example, (Ebohon et al. 2021) investigated the experiences of teachers and students with virtual/online learning during the COVID-19 lockdown in Nigeria. Their study involved 703 students and 60 teachers from five Nigerian universities and used questionnaires to examine virtual classrooms, learning outcomes, assessment, online teaching and satisfaction. The study reported substantial difficulties with Internet connectivity and challenges associated with the transition from conventional to virtual learning

Similarly, (Egielewa et al. 2022) examined online learning among 1,134 students from Nigerian universities, polytechnics and colleges of education. Their quantitative study found dissatisfaction with the virtual learning arrangements used by many institutions during the lockdown, with poor Internet infrastructure and inadequate electricity identified among important challenges. These findings demonstrate that simply introducing an online platform does not automatically create an effective e-learning environment.

(Ogolodom et al. 2023) also investigated the experiences of 540 nursing and radiography undergraduate students in Nigerian universities. The study identified financial constraints, lack of Internet access, lack of technical knowledge, poor communication with lecturers and peers, unstable Internet connectivity, high data costs and lack of appropriate devices among the challenges encountered by students. Nevertheless, many respondents regarded online learning as beneficial to their education.

These challenges are relevant to the design of an e-learning meeting system. A system designed for a Nigerian higher-education environment should recognize that students may access educational services through mobile phones, laptops and other devices under varying network conditions. It should therefore aim for a relatively simple interface, efficient communication mechanisms and organized access to meeting information and learning activities.

Existing platforms already provide some of these capabilities. For example, Microsoft Teams for Education supports virtual class meetings, direct communication, channel discussions, screen sharing, meeting scheduling, assignments, grading and class resources. Zoom provides synchronous meeting facilities, including breakout rooms that allow participants to be separated into smaller discussion groups. Learning management systems such as Moodle provide broader learning-management functionality. However, these systems are designed for broad international use and may contain extensive features that are unnecessary for a small institution, department or specific tutor-student meeting environment.

The proposed Tutor and Student E-Learning Meeting System is therefore conceived as a focused web-based system for facilitating communication and interaction between tutors and students. Rather than treating online learning merely as the electronic distribution of course materials, the proposed system emphasizes the meeting relationship between tutor and student. The system can provide user registration and authentication, role-based access, meeting scheduling, meeting information, student participation, tutor-student communication, learning-resource sharing and basic attendance or participation records. Depending on the implementation, it may also provide integration with or embedding of online meeting functionality.

The need for such a system is supported by literature showing that effective online learning depends on several interconnected factors, including technology quality, information quality, support, learner characteristics, instructor characteristics, usefulness and satisfaction. (Al-Fraihat et al. 2020), in an empirical study involving 563 students, found that technical system quality, information quality, service quality, support system quality, learner quality, instructor quality and perceived usefulness were important determinants of e-learning satisfaction and use.

Consequently, the development of a tutor and student e-learning meeting system is not simply a technological exercise. It is an attempt to apply web-based technologies to a specific educational communication problem: providing a structured environment through which tutors and students can arrange, conduct and support academic meetings without depending entirely on physical classroom attendance or scattered communication channels.

The proposed system is particularly relevant to ND/HND institutions where students and tutors increasingly use smartphones, computers and Internet-based services for academic communication. A dedicated system can centralize information that might otherwise be distributed across messaging applications, email, social-media groups and unrelated video-conferencing platforms. By providing a structured environment for tutor-student meetings, the system can contribute to improved organization, accessibility, communication and continuity of academic activities.

1.2 Statement of the Problem

Despite the increasing adoption of digital technologies in education, several problems continue to affect effective online interaction between tutors and students.

First, conventional face-to-face learning depends heavily on physical presence. Students who are unable to attend a physical class because of distance, timetable conflicts, temporary absence or other circumstances may miss important academic discussions. Although e-learning provides an alternative, not every institution has a dedicated system that organizes tutor-student meetings in a structured manner.

Second, academic communication is often fragmented across several platforms. A tutor may announce a meeting through WhatsApp, send learning materials through email, conduct a video meeting through another application and record attendance manually. Students may consequently have difficulty determining which platform to use, when a meeting will take place, how to access it and where relevant materials can be found. This fragmentation can reduce the organization and efficiency of online academic activities.

Third, the quality of tutor-student interaction may be inadequate in some online learning environments. Nigerian studies have reported poor communication with lecturers and peers as one of the challenges experienced during online learning (Ogolodom et al., 2023). A system that merely provides access to digital materials without facilitating interaction may fail to address an important part of the teaching and learning process.

Fourth, existing general-purpose platforms may provide more functions than are required for a specific educational institution or department. For example, a platform may combine meetings, corporate communication, file management, productivity tools and other services. While such integration can be beneficial, it can also introduce complexity for users who require only a straightforward tutor-student meeting environment.

Fifth, monitoring student participation can be difficult when academic meetings are organized through disconnected applications. Without a central system, maintaining records of scheduled meetings, participants, attendance and related academic activities may require considerable manual effort.

Sixth, research has established that Internet connectivity, electricity, device availability, cost and technical skills remain significant considerations in Nigerian e-learning environments (Ebohon et al., 2021; Egielewa et al., 2022; Ogolodom et al., 2023). Therefore, an appropriate solution should be designed with simplicity and accessibility in mind rather than assuming ideal technological conditions.

The problem addressed by this project is therefore the absence or inadequacy of a focused system that brings tutor-student meeting scheduling, communication, participation and related academic information into one organized environment. The proposed system seeks to address this problem by developing a web-based platform that provides structured interaction between tutors and students.

1.3 Aim of the Study

The aim of this study is to design and implement a Tutor and Student E-Learning Meeting System that facilitates organized online communication, meeting scheduling, participation and academic interaction between tutors and students.

1.4 Objectives of the Study

The specific objectives of the study are to:

- Examine the existing methods used by tutors and students to organize and conduct academic meetings.
- Identify the major challenges associated with existing tutor-student online meeting and communication methods.
- Design a user-friendly web-based e-learning meeting system for tutors and students.
- Develop a secure user authentication mechanism for tutors and students.
- Implement a meeting scheduling module that enables tutors to create and manage academic meetings.

- Provide students with the ability to view available meetings and participate in scheduled academic sessions.
- Provide communication facilities that support interaction between tutors and students.
- Provide facilities for tutors to share relevant academic information or learning resources with students.
- Provide basic records of scheduled meetings and student participation.
- Evaluate the proposed system in terms of functionality, usability and its ability to improve the organization of tutor-student online interaction.

1.5 Research Questions

The study seeks to answer the following research questions:

1. What methods are currently used by tutors and students to organize academic meetings?
2. What problems are associated with existing methods of tutor-student online communication and meetings?
3. What features are required in an effective tutor and student e-learning meeting system?
4. How can a web-based system improve the organization of tutor-student academic meetings?
5. How can the proposed system facilitate effective communication between tutors and students?
6. How can the system support meeting scheduling and student participation?
7. How usable and acceptable is the proposed system to its intended users?

1.6 Significance of the Study

The study is significant to tutors, students, educational institutions, system developers and researchers.

1.6.1 Significance to Tutors

The system can provide tutors with a structured method of creating and managing academic meetings. Instead of relying on several unrelated communication channels, tutors can use a centralized platform to provide meeting information, communicate with students and organize learning activities. The system can also reduce the administrative effort associated with manually coordinating meetings.

1.6.2 Significance to Students

Students can benefit from having a centralized platform through which they can view scheduled meetings, access relevant information and communicate with tutors. This can reduce uncertainty regarding meeting dates, times and access procedures. The system may also provide students with greater flexibility in participating in academic activities when physical attendance is difficult.

1.6.3 Significance to Educational Institutions

Educational institutions can use the proposed system as a model for supporting digital learning and academic communication. A dedicated platform may help institutions organize online academic activities while maintaining appropriate records of meetings and participation.

1.6.4 Significance to System Developers

The project provides a practical example of applying web-development concepts, database management, user authentication, role-based access control, interface design and online communication technologies to an educational problem.

1.6.5 Significance to Researchers

The study can provide a basis for further research into online learning systems, student engagement, tutor-student interaction, virtual classrooms, learning management systems and educational technology in Nigerian institutions.

1.7 Scope of the Study

The study focuses on the design and implementation of a web-based Tutor and Student E-Learning Meeting System.

- The proposed system covers:
- User registration and authentication.
- Separate tutor and student roles.
- Tutor profile and student profile management.
- Creation and scheduling of academic meetings.
- Viewing of scheduled meetings.
- Meeting details and access information.

- Tutor-student communication.
- Sharing of relevant learning information or resources.
- Recording or monitoring of basic meeting participation/attendance.
- Administrative management of users and system information.
- Database storage and retrieval of relevant records.

The study focuses primarily on the academic interaction between tutors and students. It does not attempt to develop a complete institutional Learning Management System covering every possible function such as comprehensive examination management, school fees management, student admission, hostel management or institutional payroll.

Where live video conferencing is required, the system may use an appropriate online meeting service or meeting technology rather than attempting to reproduce the complete infrastructure of a large commercial video-conferencing provider.

1.8 Limitations of the Study

The study may be affected by several limitations.

First, Internet connectivity can influence the performance of an online meeting system. Students and tutors with slow or unstable connections may experience difficulty accessing real-time services.

Second, electricity supply can affect access to computers, mobile devices and network equipment. This is particularly relevant in environments where power supply is unreliable.

Third, the cost of Internet data and compatible devices may limit participation by some students. Nigerian studies have identified Internet access, device availability and financial constraints as significant challenges to online learning (Ogolodom et al., 2023; Egielewa et al., 2022).

Fourth, users may have different levels of digital literacy. Some students or tutors may require orientation before they can use all functions of the system effectively.

Fifth, the scope of an undergraduate project limits the extent to which the proposed system can reproduce the advanced infrastructure, scalability and security mechanisms of large commercial platforms.

Finally, the evaluation of the system may be conducted using a relatively small number of users. Consequently, the findings may not be generalized to every higher-education institution without further research.

1.9 Definition of Key Terms

E-Learning

E-learning refers to the use of information and communication technologies to facilitate, support, deliver, and manage teaching and learning activities through electronic or digital environments. It encompasses different forms of online and technology-supported learning, including virtual learning environments, web-based learning, and collaborative digital learning. (Giannakos et al. 2022)

Online Learning

Online learning is a form of education in which teaching, learning materials, communication, and interaction are facilitated through internet-connected digital technologies. (Martin et al. 2020)

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Tutor

A tutor is an individual who provides instruction, academic guidance, explanation, supervision, or learning support to students. Within an online learning environment, the tutor's role may extend beyond delivering content to include facilitating discussions, communicating with learners, providing feedback, and managing learning activities (Martin et al., 2020).

Student

A student is a learner who participates in educational activities with the purpose of acquiring knowledge, skills, competencies, or qualifications. In technology-supported learning environments, students interact with instructors, course materials, communication tools, and learning platforms as part of the learning process (Rodrigues et al., 2019)

E-Learning Meeting

An e-learning meeting is a planned online interaction between tutors and students conducted through digital communication technologies for teaching, discussion, consultation, supervision, or other academic purposes. Synchronous online learning allows participants to interact with instructors and other learners at approximately the same time through technologies such as video conferencing (Mpungose, 2021).

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Synchronous Learning

Synchronous learning is a form of online learning in which instructors and learners participate in educational activities at the same time, even when they are physically located in different places. (Regmi and Jones, 2020)

Asynchronous Learning

Asynchronous learning refers to learning activities that do not require tutors and students to participate at the same time. Learners may access learning materials, messages, assignments, or other academic resources at different times according to their availability. (Regmi and Jones, 2020)

Learning Management System (LMS)

A Learning Management System (LMS) is a software-based platform designed to support the administration, delivery, management, monitoring, and organization of learning activities. LMS platforms commonly provide functions for course materials, communication, assessment, learner activities, and tracking. (Turnbull et al, 2021)

Video Conferencing

Video conferencing is a communication technology that enables people in different physical locations to communicate through real-time audio and video over a network. In education, video conferencing can support synchronous interaction between students and instructors. (Edumadze et al, 2022)

Virtual Classroom

A virtual classroom is a digital learning environment in which instructors and learners interact through internet-based technologies to conduct teaching and learning activities. It can incorporate synchronous communication, learning materials, discussion, and other educational activities. (Regmi and Jones, 2020)

User Authentication

User authentication is the process of verifying the identity of a person attempting to access a computer system or application. It commonly involves credentials such as usernames, passwords, or other identity-verification mechanisms.

Student Engagement

Student engagement refers to the extent to which students actively participate in and interact with learning activities. In digital learning environments, engagement can be reflected through activities such as accessing course materials, participating in communication, completing assessments, and interacting with learning platforms. (Bedenlier et al. 2023)

Meeting Scheduling

Meeting scheduling refers to the process of planning and organizing a meeting by specifying relevant information such as the date, time, participants, and purpose of the meeting.

Digital Learning Environment

A digital learning environment is a technology-supported environment in which learning resources, communication, interaction, and educational activities are provided through digital technologies. (Giannakos et al. (2022)

Learning Resources

Learning resources are materials or information used to support teaching and learning. They may include documents, lecture notes, presentations, videos, links, assignments, and other educational materials. E-learning environments are particularly useful for providing learners with online access to educational resources (Regmi & Jones, 2020).

E-Learning Platform

An e-learning platform is a digital software environment designed to facilitate one or more aspects of electronic education, including content delivery, communication, collaboration, assessment, or learner management. (Rodrigues et al., 2019; Giannakos et al., 2022).

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature relevant to the design and implementation of the Tutor and Student E-Learning Meeting System. The review examines the concepts, technologies, theories, empirical studies and existing systems that provide a foundation for the proposed system.

The review is organized into seven major sections: conceptual review, theoretical review, empirical review, review of related/existing systems, summary of the literature review and research gap. Particular attention is given to online learning, tutor-student interaction, synchronous learning, learning management systems, student engagement, virtual meetings, system usability, technology acceptance and the Nigerian higher-education environment.

The literature shows that the effectiveness of e-learning depends not only on the availability of technology but also on the quality of interaction, instructional design, system quality, instructor participation, student engagement and user acceptance. Recent systematic reviews have similarly emphasized adoption, readiness, interaction, engagement and system quality as important areas in e-learning research (Baig et al., 2022; Nkomo et al., 2021).

2.2 Conceptual Review

2.2.1 Concept of E-Learning

E-learning refers to the use of electronic technologies to support teaching and learning. It encompasses a wide range of activities, including online lectures, electronic learning materials, discussion forums, online assessments, communication systems, learning management systems and virtual meetings.

Modern e-learning is not limited to simply placing printed lecture notes online. It involves an interaction between learners, instructors, content and technological systems. (Baig et al, 2022), after reviewing 47 studies on e-learning in higher education, reported that the literature has addressed adoption, acceptance, readiness, user perspectives, expansion and challenges of e-learning.

E-learning can be classified into synchronous and asynchronous forms. Synchronous e-learning requires learners and instructors to participate at approximately the same time, whereas asynchronous learning allows participants to access materials and communicate according to their individual schedules.

For the present study, e-learning is considered primarily as an environment through which tutors and students can communicate and participate in organized academic meetings.

2.2.2 Synchronous E-Learning

Synchronous e-learning involves real-time interaction between instructors and learners. Examples include live online lectures, virtual tutorials, video conferences, live question-and-answer sessions and online group discussions.

One major advantage of synchronous learning is immediacy. Students can ask questions and receive responses without waiting for a later communication cycle. Tutors can observe student participation and modify explanations based on students' responses.

However, synchronous learning also has limitations. Participants need suitable devices, Internet connectivity and appropriate schedules. Poor connectivity may interrupt communication, while differences in schedules may make participation difficult.

A systematic review of synchronous online learning found that although synchronous online learning provides opportunities for continued teaching and learning, maintaining students' interest and engagement remains an important concern (Synchronous Online Learning During Movement Control Order, 2021).

2.2.3 Asynchronous E-Learning

Asynchronous e-learning allows learners and tutors to interact without being online simultaneously. Examples include discussion forums, recorded lectures, electronic documents, announcements, email and assignment submission systems.

The principal advantage is flexibility. Students can access learning resources at a time that is convenient to them. However, asynchronous learning may not provide the same level of immediate feedback as synchronous learning.

An effective e-learning environment may therefore combine both approaches. A tutor and student meeting system can use synchronous meetings for real-time interaction while supporting asynchronous activities through announcements, resources, messages and meeting information.

2.2.4 Tutor-Student Interaction

Tutor-student interaction refers to communication and academic exchange between an instructor and a learner. It includes asking and answering questions, explaining concepts, giving feedback, providing guidance and discussing academic activities.

Tutor-student interaction is particularly important in online education because physical proximity between instructors and students is reduced. (Miao et al. 2022) found that teacher-student interaction significantly affected social presence and learning engagement among undergraduate students in online learning environments.

The proposed system therefore places tutor-student interaction at the center of its design. The system is intended not simply to deliver information but to provide an organized channel for tutors and students to communicate around scheduled academic activities.

2.2.5 Student Interaction

Student interaction involves communication and collaboration among learners. It may occur through discussion forums, group meetings, collaborative assignments, chat facilities and breakout sessions.

Interaction between students can support peer learning and the development of social presence. (Miao et al. 2022) found that student interaction, like teacher-student interaction, contributed to social presence and learning engagement.

Consequently, a well-designed online meeting environment should not necessarily restrict interaction to tutor-to-student communication. Where appropriate, students should also be able to participate in group discussions and collaborative activities.

2.2.6 Virtual Classroom

A virtual classroom is an online environment designed to reproduce or support activities normally performed in a physical classroom. It can include video and audio communication, text chat, screen sharing, digital whiteboards, file sharing, attendance mechanisms and discussion facilities.

A virtual classroom differs from a simple video call because it is designed around an educational purpose. It may include course information, learning materials, assessment activities, participant management and communication.

The proposed system adopts the concept of a virtual classroom in a focused manner by concentrating on the organization of tutor-student meetings.

2.2.7 Learning Management System

A Learning Management System (LMS) is software used to manage educational activities. Common LMS functions include user management, course management, content distribution, assignments, assessments, grading, communication and activity monitoring.

(Ahmadi et al, 2023) found that LMS-based student engagement can be assessed through indicators involving system usage, access to course materials, assignments, assessments and communication. Their review of 32 studies identified 27 indicators of student engagement.

The proposed system is not intended to replace a complete LMS. Instead, it represents a specialized educational system concentrating on tutor-student meetings and associated communication.

2.2.8 Online Meeting Scheduling

Meeting scheduling is the process of specifying the details of an online academic meeting. Such details may include:

- Meeting title.
- Date.
- Start time.
- End time.
- Tutor.
- Intended participants.
- Course or subject.
- Meeting description.
- Meeting access information.
- Relevant learning materials

A scheduling mechanism is important because it changes an online meeting from an informal communication event into a structured academic activity.

2.2.9 Online Communication

Online communication refers to the exchange of information through networked digital technologies. Within an e-learning system, communication can be synchronous or asynchronous.

Examples include:

- Text messaging.
- Discussion forums.
- Announcements.
- Email notifications.
- Audio communication.
- Video communication.
- File sharing.

Microsoft Teams for Education demonstrates how meetings, chats, channel posts and educational activities can be integrated into one environment.

The proposed system applies the same general principle of centralizing relevant communication, but within the narrower context of tutor-student academic meetings.

2.2.10 Student Engagement

Student engagement describes students' active involvement in learning. It includes behavioural participation, cognitive involvement and emotional involvement.

(Nkomo et al. 2021) found that student engagement with digital technologies is a complex concept that requires consideration of how students interact with technology and learning activities.

(Limbu and McKinley, 2025) conducted a systematic review of 50 studies and identified student-related, instructor-related, course-design-related and technology/system-related factors associated with online student engagement. Instructor-student interaction, teaching presence, course organization, learning activities, technology and perceived usefulness were among the factors identified.

For the proposed project, student engagement can be supported through scheduled meetings, communication, participation records and access to relevant academic information.

2.2.11 Learning Resources

Learning resources are materials used to support teaching and learning. They can include:

- Lecture notes.
- PDF documents.
- Presentations.
- Videos.
- Images.
- Links to educational websites.
- Assignments.
- Supplementary reading materials.

A tutor-student meeting system can provide a location for linking meeting activities with relevant resources. This helps students understand the academic purpose of a scheduled meeting.

2.2.12 User Authentication

Authentication is the process of verifying the identity of a system user. In the proposed system, users may log in using credentials such as an email address or username and password.

Authentication is important because the system may contain academic information and communication intended only for authorized users.

2.2.13 Authorization and Role-Based Access Control

Authorization determines what an authenticated user can do within a system. Role-based access control assigns permissions according to roles.

For example:

- User Role| Possible Permissions
- Administrator| Manage users, system records and settings

- Tutor| Create meetings, communicate with students, provide resources and manage academic sessions
- Student| View meetings, join sessions, communicate with tutors and access resources
- Role-based access control prevents users from accessing functions outside their responsibilities and provides a structured security model for the system.

2.2.14 Database Management

A database is an organized collection of related information. The proposed system may store information such as:

- User accounts.
- Tutor profiles.
- Student profiles.
- Courses or subjects.
- Meeting records.
- Attendance records.
- Messages.
- Learning resources.

A relational database can organize these records into related tables. For example, a "Users" table can be connected to "Meetings", while a "Meetings" table can be connected to "Attendance".

2.2.15 Web-Based Application

A web-based application is software accessed through a web browser. It normally consists of a front-end interface, application logic, database and server-side components.

The proposed system may use technologies such as:

- HTML for page structure.
- CSS for presentation and responsive design.
- JavaScript for client-side interaction.
- A server-side programming language such as PHP, Python, Java or Node.js.

- MySQL or another relational database.
- web server for hosting the application.

The exact technology stack depends on the implementation adopted for the project.

2.2.16 Responsive Web Design

Responsive web design enables a website to adapt its layout to different screen sizes. This is particularly important for an e-learning system because students may access the platform through smartphones, tablets and computers.

A responsive interface can improve usability by ensuring that navigation, meeting information, buttons and forms remain usable across different devices.

2.2.17 Usability

Usability refers to how easily users can learn and use a system to achieve their intended goals. An e-learning meeting system should have clear navigation, understandable labels, consistent interfaces and simple procedures for scheduling and joining meetings.

Poor usability can discourage users even when the underlying technology is technically functional. This is important because e-learning success is influenced by both technical and human factors.

2.2.18 System Quality

System quality refers to characteristics such as reliability, usability, accessibility, responsiveness, security and technical performance.

(Al-Fraihat et al, 2020) identified technical system quality, information quality, service quality, support system quality, learner quality, instructor quality and perceived usefulness among important factors associated with e-learning success.

This suggests that evaluating the proposed system should go beyond asking whether individual functions work. The system should also be considered in terms of usability, usefulness and reliability.

2.3 Theoretical Review

2.3.1 Community of Inquiry Framework

The Community of Inquiry (CoI) framework developed by Garrison, Anderson and Archer is one of the most relevant theoretical frameworks for this project. The original framework explains meaningful online learning through the interaction of three forms of presence: teaching presence, social presence and cognitive presence (Garrison et al., 2000).

Teaching Presence

Teaching presence refers to the design, organization, facilitation and direction of learning activities by the instructor. It includes preparing learning activities, communicating expectations, facilitating discussion and providing direct instruction.

For the proposed system, teaching presence can be supported by allowing tutors to schedule meetings, provide meeting descriptions, share learning resources and communicate with students.

Social Presence

Social presence concerns the ability of participants to present themselves as real people and develop meaningful relationships within the online environment. Communication facilities, group discussions and synchronous meetings can contribute to social presence.

This is highly relevant to the proposed system because tutor-student meetings provide opportunities for direct communication and interaction.

Cognitive Presence

Cognitive presence relates to learners' ability to construct and confirm meaning through reflection, discussion and inquiry.

A tutor and student meeting system can support cognitive presence by providing opportunities for tutors to explain concepts, ask questions, facilitate discussions and provide feedback.

(Caskurlu et al., 2021), in a thematic synthesis of 35 qualitative studies informed by the CoI framework, identified course design, instructor actions and student actions as major categories influencing online learning experiences.

(Martin et al, 2022) conducted a meta-analysis of 19 empirical studies and examined relationships between CoI presences and learning outcomes, including actual learning, perceived learning and satisfaction.

The CoI framework is therefore appropriate because the proposed system is specifically concerned with facilitating the interaction among tutors, students and learning activities.

2.3.2 Transactional Distance Theory

Moore's Transactional Distance Theory provides another useful theoretical foundation. The theory explains that distance education involves more than physical separation. Transactional distance is influenced by dialogue, structure and learner autonomy.

Dialogue represents communication between learner and instructor. Structure refers to the organization and flexibility of instructional activities, while learner autonomy concerns the learner's ability to manage aspects of learning independently.

This theory is directly relevant to the proposed system because the system is intended to reduce communication difficulties between tutors and students. A scheduled meeting system increases opportunities for dialogue while providing structure for online academic activities.

The theory also supports the importance of tutor-student interaction in distance learning. An online system that provides only static content may leave students with limited opportunities to obtain clarification or feedback. A meeting-oriented system can reduce this problem by creating organized opportunities for dialogue.

2.3.3 Technology Acceptance Model

The Technology Acceptance Model (TAM), proposed by (Davis, 1989), explains users' acceptance of information technology through concepts including perceived usefulness and perceived ease of use. Davis found that these constructs were important predictors of user acceptance of information technology.

Perceived usefulness refers to the degree to which a user believes that using a system will improve task performance. In the proposed project, students and tutors are more likely to accept the system if they believe it makes meetings easier to organize and participate in.

Perceived ease of use refers to the degree to which a user believes that using the system will require little effort. Therefore, complicated registration procedures, confusing navigation or difficult meeting-access processes could negatively affect acceptance.

(Cavus et al, 2022), in a systematic literature review of TAM and learning management systems, highlighted the continued relevance of technology acceptance research to LMS adoption in higher education.

The TAM therefore provides a useful framework for evaluating the usability and acceptability of the proposed system.

2.3.4 Theoretical Framework for the Proposed Study

The three theories complement one another:

Theory/Framework	Main Concern	Relevance to Proposed System
Community of Inquiry	Teaching, social and cognitive presence	Guides interaction and online learning design
Transactional Distance Theory	Dialogue, structure and learner autonomy	Supports organized tutor-student communication
Technology Acceptance Model	Usefulness and ease of use	Guides system usability and acceptance

The Community of Inquiry framework is adopted as the principal theoretical foundation because the central purpose of the proposed system is to facilitate meaningful interaction between tutors and students. Transactional Distance Theory provides additional justification for structured communication, while TAM provides a basis for considering whether users will accept and use the developed system.

2.4 Empirical Review

2.4.1 (Baig, Shuib and Yadegaridehkordi, 2022)

(Baig et al., 2022) conducted a systematic literature review on e-learning adoption in higher education. The researchers reviewed 47 relevant studies published between 2011 and 2019. Their purpose was to classify research according to areas of focus, theoretical models and research methods.

The review found that existing studies addressed e-learning adoption, acceptance, readiness, user perspectives, expansion and challenges. The researchers also observed that many studies used

quantitative approaches and that theoretical frameworks were frequently applied during post-adoption stages.

The study is relevant to the present project because it demonstrates that successful e-learning development requires consideration of adoption, user experience and implementation challenges rather than focusing only on software functionality.

2.4.2 (Al-Fraihat, Joy, Masa'deh and Sinclair, 2020)

(Al-Fraihat et al, 2020) conducted an empirical study evaluating e-learning system success in a UK university. The study used data from 563 students and applied Partial Least Squares Structural Equation Modelling (PLS-SEM).

The researchers developed a comprehensive model integrating elements of the DeLone and McLean information systems success model, Technology Acceptance Model and e-learning quality models. The findings identified technical system quality, information quality, service quality, support system quality, learner quality, instructor quality and perceived usefulness as important determinants of satisfaction.

The study is relevant because it demonstrates that the success of an e-learning system depends on technical and human factors. The proposed project should therefore consider usability, usefulness, system quality and instructor participation.

2.4.3 (Nkomo, Daniel and Butson, 2021)

(Nkomo et al., 2021) conducted a systematic review examining student engagement with digital technologies. The study synthesized existing literature to understand how digital technologies influence student engagement.

The researchers emphasized the complexity of student engagement and the need to consider how learners interact with digital learning technologies.

The study is relevant to the present project because a tutor-student meeting system should be designed not merely as a communication tool but as an environment capable of supporting meaningful student participation.

2.4.4 (Miao, Chang and Ma, 2022)

(Miao et al, 2022) investigated the effects of teacher-student interaction, student-student interaction and social presence on learning engagement in online learning. The researchers collected data from 354 full-time undergraduate students in a large public university in China.

The study found that teacher-student interaction and student-student interaction directly influenced social presence and learning engagement. Social presence also directly influenced engagement and mediated the relationships between interaction and engagement.

This study is particularly relevant because tutor-student interaction is the central focus of the proposed system. It provides empirical support for designing a system that facilitates direct communication rather than concentrating exclusively on content delivery.

2.5 Review of Related/Existing Systems

Several existing technologies support online teaching, communication and meetings. Their examination is important because the proposed system should build on established concepts while addressing limitations relevant to the project's intended environment.

2.5.1 Zoom

Zoom is a widely used video-conferencing platform that supports synchronous communication. It provides features such as audio and video meetings, screen sharing, chat and breakout rooms. Breakout rooms allow participants to be divided into smaller groups for focused discussions. Zoom provides useful meeting technology, but a specialized academic system may be required to organize the broader tutor-student meeting process.

2.5.2 Microsoft Teams for Education

Microsoft Teams for Education integrates virtual meetings with educational communication and class-management functions. Its education features include class meetings, chat, channel posts, assignments, grading, resources and other collaboration functions.

Configuration and institutional account requirements may complicate adoption in some settings. The proposed system differs by concentrating specifically on tutor-student meetings and related academic interaction.

2.5.3 Learning Management Systems

LMS platforms provide broader educational management features, including course content, assignments, assessments, communication and student monitoring. They are appropriate when an institution requires a complete online learning environment.

A complete LMS may be excessive for an institution primarily seeking tutor-student meeting functionality.

Real-time video meetings may require integration with separate services.

Implementation may require significant technical and administrative resources.

The proposed system is narrower and focuses on meeting coordination and communication.

2.5.4 WhatsApp and Similar Messaging Platforms

Messaging applications are frequently used informally by tutors and students to exchange announcements, messages, documents and meeting information.

It is not specifically designed for institutional academic management.

The proposed system can provide a more structured environment than informal messaging applications.

2.6 Summary of the Literature Review

The literature reviewed in this chapter demonstrates that e-learning has become an important component of higher education. However, effective e-learning requires more than providing students with electronic materials.

The conceptual review established that e-learning involves several related concepts, including synchronous and asynchronous learning, tutor-student interaction, student engagement, virtual classrooms, LMSs, online communication, meeting scheduling, authentication, authorization, databases and responsive web applications.

The theoretical review identified the Community of Inquiry framework as the principal theoretical foundation for the study. The framework emphasizes teaching presence, social presence and cognitive presence. Transactional Distance Theory further emphasizes dialogue, structure and learner autonomy, while the Technology Acceptance Model highlights perceived usefulness and ease of use.

The empirical studies reviewed show that tutor-student interaction is associated with student engagement and social presence (Miao et al., 2022). Research has also demonstrated that e-

learning system success depends on technical quality, information quality, service quality, learner characteristics, instructor characteristics and perceived usefulness (Al-Fraihat et al., 2020).

The Nigerian studies reviewed reveal additional challenges. Poor Internet connectivity, electricity problems, device limitations, cost, technical skills and communication difficulties have affected online learning experiences in Nigerian higher education (Ebohon et al., 2021; Egielewa et al., 2022; Ogolodom et al., 2023).

The review of existing systems shows that platforms such as Zoom and Microsoft Teams already provide powerful virtual meeting and communication facilities. However, these systems are general-purpose or broad educational platforms. A focused tutor-student meeting system can provide a simpler and more institution-specific environment for scheduling, communication, participation and meeting management.

2.7 Research Gap

The reviewed literature reveals several gaps that justify the development of the proposed Tutor and Student E-Learning Meeting System.

First, a substantial proportion of existing research focuses on the adoption, acceptance, effectiveness and challenges of e-learning, rather than on developing a focused system specifically for tutor-student meeting management. (Miao et al, 2022), for example, found that much of the literature focuses on adoption, acceptance, readiness and user experiences.

Second, although studies have established the importance of tutor-student interaction, interaction is frequently investigated as a variable influencing engagement rather than as the central design objective of an educational software system. (Miao et al, 2022) demonstrated that teacher-student interaction significantly affects engagement and social presence, but the study did not provide a dedicated tutor-student meeting management system.

Third, existing commercial platforms such as Zoom and Microsoft Teams provide extensive communication and collaboration features, but their broad functionality may exceed the requirements of a focused academic meeting environment. The proposed system seeks to concentrate on the core processes of tutor-student meeting management rather than reproducing every feature of a large commercial platform.

Fourth, much of the available empirical evidence on online learning comes from countries and educational environments with technological and infrastructural conditions different from those found in Nigeria. Nigerian studies have repeatedly identified connectivity, electricity, cost, device

availability and technical knowledge as important challenges (Ebohon et al., 2021; Egielewa et al., 2022; Ogolodom et al., 2023). There is therefore a need for project-level systems that are designed with the practical conditions of Nigerian students and tutors in mind.

Fifth, existing research increasingly recognizes student engagement as multidimensional. (Limbu and McKinley 2025) identified student, instructor, course-design and technology-related factors influencing engagement. However, there remains a practical need to translate these findings into a relatively simple software environment that supports structured tutor interaction, student participation and meeting organization.

Sixth, LMS research has identified system usage, access to learning materials, assignments, assessments and communication as useful indicators of engagement (Ahmadi et al., 2023). Nevertheless, a focused system that connects these activities specifically to scheduled tutor-student meetings can provide a more direct relationship between academic interaction and system records.

The research gap can therefore be summarized as follows:

Existing research establishes the importance of e-learning, system quality, student engagement and tutor-student interaction, while existing platforms provide meeting and communication technologies. However, there remains a practical gap in the development of a focused, simple and customizable system that specifically organizes tutor-student academic meetings, communication, participation and related learning information within one environment, particularly with the practical challenges of Nigerian higher education in mind.

The proposed Tutor and Student E-Learning Meeting System seeks to address this gap by translating the findings of existing research into a practical web-based system. Its design is informed by the Community of Inquiry framework, Transactional Distance Theory and Technology Acceptance Model, while its functional requirements are guided by empirical evidence concerning interaction, engagement, system quality and Nigerian e-learning challenges.